

Sheth L.U.J & Sir M.V. College Of Science
Subject :- Cyber & Information Security
Practical no 5

Code :-

```
print("----- Diffie-Hellman Key Exchange -----")
```

```
p = int(input("Enter prime number (p): "))
```

```
g = int(input("Enter primitive root (g): "))
```

```
a = int(input("Enter Alice's private key: "))
```

```
b = int(input("Enter Bob's private key: "))
```

```
A = pow(g, a, p)
```

```
B = pow(g, b, p)
```

```
print("\nAlice's Public Key:", A)
```

```
print("Bob's Public Key:", B)
```

```
alice_secret = pow(B, a, p)
```

```
bob_secret = pow(A, b, p)
```

```
print("\nAlice's Shared Secret:", alice_secret)
```

```
print("Bob's Shared Secret:", bob_secret)
```

```
if alice_secret == bob_secret:
```

```
    print("\nKey Exchange Successful!")
```

```
    print("Both parties have the same shared secret key.")
```

```
else:
```

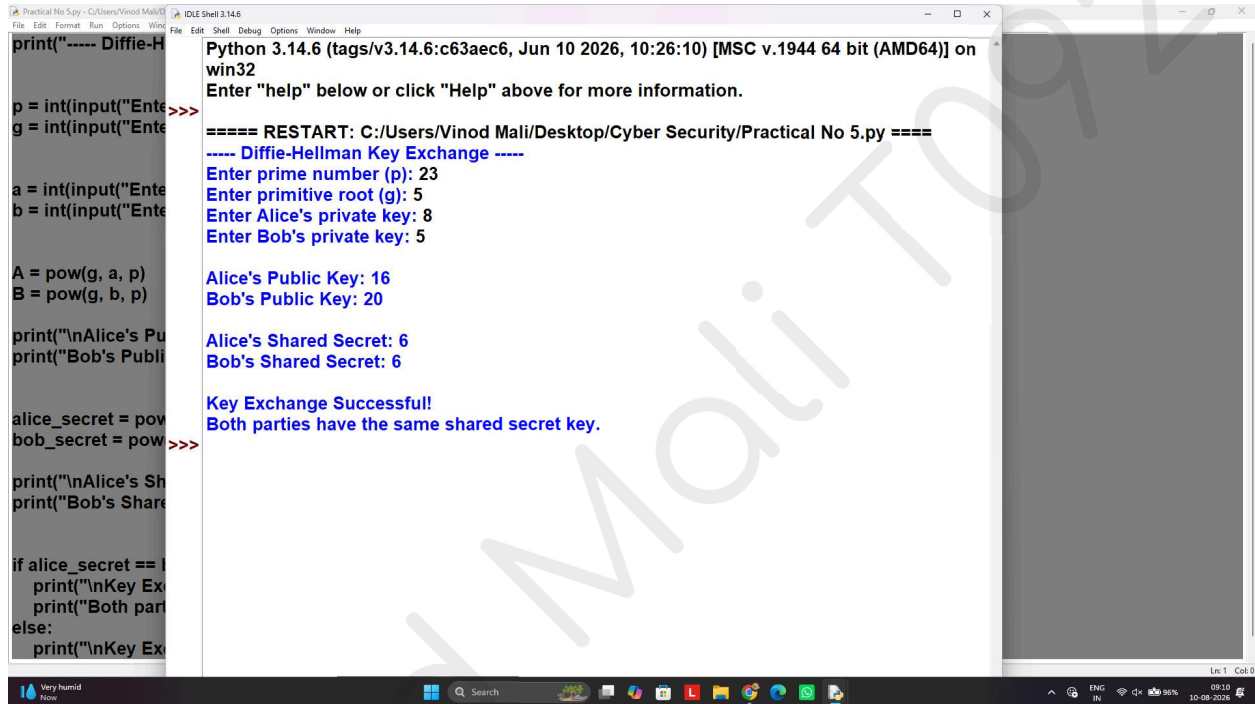
```
    print("\nKey Exchange Failed!")
```

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Output :-



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/Vinod Mali/Desktop/Cyber Security/Practical No 5.py =====
----- Diffie-Hellman Key Exchange -----
Enter prime number (p): 23
Enter primitive root (g): 5
Enter Alice's private key: 8
Enter Bob's private key: 5

Alice's Public Key: 16
Bob's Public Key: 20

Alice's Shared Secret: 6
Bob's Shared Secret: 6

Key Exchange Successful!
Both parties have the same shared secret key.
```

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Code :- GUI

```
import tkinter as tk
from tkinter import messagebox
```

```
def calculate_keys():
```

```
    try:
```

```
        p = int(p_entry.get())
```

```
        g = int(g_entry.get())
```

```
        a = int(alice_private_entry.get())
```

```
        b = int(bob_private_entry.get())
```

```
        if p <= 1 or g <= 0 or a <= 0 or b <= 0:
```

```
            messagebox.showerror("Error", "Enter valid positive numbers.")
```

```
            return
```

```
        A = pow(g, a, p)
```

```
        B = pow(g, b, p)
```

```
        alice_secret = pow(B, a, p)
```

```
        bob_secret = pow(A, b, p)
```

```
        alice_public_label.config(text=f"Public Key: {A}")
```

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```
bob_public_label.config(text=f"Public Key: {B}")
```

```
alice_secret_label.config(text=f"Shared Secret: {alice_secret}")
```

```
bob_secret_label.config(text=f"Shared Secret: {bob_secret}")
```

```
if alice_secret == bob_secret:
    result_label.config(
        text="✓ KEY EXCHANGE SUCCESSFUL",
        fg="#00ff66"
    )
else:
    result_label.config(
        text="X KEY EXCHANGE FAILED",
        fg="#ff3333"
    )
```

```
except ValueError:
    messagebox.showerror(
        "Invalid Input",
        "Please enter numbers only."
    )
```

```
def clear_fields():
    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)
    alice_private_entry.delete(0, tk.END)
    bob_private_entry.delete(0, tk.END)
```

```
alice_public_label.config(text="Public Key: ---")
bob_public_label.config(text="Public Key: ---")
```

```
alice_secret_label.config(text="Shared Secret: ---")
bob_secret_label.config(text="Shared Secret: ---")
```

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```
result_label.config(text="")
```

```
root = tk.Tk()
root.title("Diffie-Hellman | Secure Key Exchange")
root.geometry("750x550")
root.configure(bg="#080808")
root.resizable(False, False)
```

```
GREEN = "#00ff66"
DARK_GREEN = "#003b1f"
WHITE = "#eeeeee"
GRAY = "#999999"
BLACK = "#080808"
```

```
tk.Label(
    root,
    text="> DIFFIE-HELLMAN KEY EXCHANGE",
    font=("Consolas", 20, "bold"),
    bg=BLACK,
    fg=GREEN
).pack(pady=(20, 5))
```

```
tk.Label(
    root,
    text="SECURE COMMUNICATION PROTOCOL",
    font=("Consolas", 10),
    bg=BLACK,
    fg=GRAY
).pack(pady=(0, 20))
```

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```
param_frame = tk.Frame(  
    root,  
    bg="#101010",  
    highlightbackground=GREEN,  
    highlightthickness=1  
)  
param_frame.pack(padx=30, fill="x")  
  
tk.Label(  
    param_frame,  
    text="[ PUBLIC PARAMETERS ]",  
    font=("Consolas", 11, "bold"),  
    bg="#101010",  
    fg=GREEN  
)grid(row=0, column=0, columnspan=4, pady=10)  
  
tk.Label(  
    param_frame,  
    text="Prime ( $\rho$ ):",  
    font=("Consolas", 10),  
    bg="#101010",  
    fg=WHITE  
)grid(row=1, column=0, padx=10, pady=10)  
  
 $\rho$ _entry = tk.Entry(  
    param_frame,  
    width=15,  
    bg="#050505",  
    fg=GREEN,  
    insertbackground=GREEN,  
    font=("Consolas", 11),  
    relief="flat"  
)  
 $\rho$ _entry.grid(row=1, column=1, padx=10)
```

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```
tk.Label(  
    param_frame,  
    text="Generator (g):",  
    font=("Consolas", 10),  
    bg="#101010",  
    fg=WHITE  
)grid(row=1, column=2, padx=10)  
  
g_entry = tk.Entry(  
    param_frame,  
    width=15,  
    bg="#050505",  
    fg=GREEN,  
    insertbackground=GREEN,  
    font=("Consolas", 11),  
    relief="flat"  
)  
g_entry.grid(row=1, column=3, padx=10)  
  
users_frame = tk.Frame(root, bg=BLACK)  
users_frame.pack(pady=25)  
  
alice_frame = tk.Frame(  
    users_frame,  
    bg="#101010",  
    width=300,  
    height=180,  
    highlightbackground=GREEN,  
    highlightthickness=1  
)  
alice_frame.grid(row=0, column=0, padx=10)  
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```
tk.Label(  
    alice_frame,  
    text="ALICE",  
    font=("Consolas", 15, "bold"),  
    bg="#101010",  
    fg=GREEN  
).pack(pady=10)
```

```
tk.Label(  
    alice_frame,  
    text="Private Key:",  
    font=("Consolas", 10),  
    bg="#101010",  
    fg=WHITE  
).pack()
```

```
alice_private_entry = tk.Entry(  
    alice_frame,  
    width=25,  
    bg="#050505",  
    fg=GREEN,  
    insertbackground=GREEN,  
    font=("Consolas", 10),  
    relief="flat"  
)  
alice_private_entry.pack(pady=5)
```

```
alice_public_label = tk.Label(  
    alice_frame,  
    text="Public Key: ---",  
    font=("Consolas", 10),  
    bg="#101010",  
    fg=GREEN
```

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```
)  
alice_public_label.pack(pady=5)
```

```
alice_secret_label = tk.Label(  
    alice_frame,  
    text="Shared Secret: ---",  
    font=("Consolas", 10, "bold"),  
    bg="#101010",  
    fg=GREEN  
)  
alice_secret_label.pack(pady=5)
```

```
bob_frame = tk.Frame(  
    users_frame,  
    bg="#101010",  
    width=300,  
    height=180,  
    highlightbackground=GREEN,  
    highlightthickness=1  
)  
bob_frame.grid(row=0, column=1, padx=10)
```

```
tk.Label(  
    bob_frame,  
    text="BOB",  
    font=("Consolas", 15, "bold"),  
    bg="#101010",  
    fg=GREEN  
) .pack(pady=10)
```

```
tk.Label(  
    bob_frame,  
    text="Private Key:",
```

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```
font=("Consolas", 10),  
bg="#101010",  
fg=WHITE  
).pack()
```

```
bob_private_entry = tk.Entry(  
    bob_frame,  
    width=25,  
    bg="#050505",  
    fg=GREEN,  
    insertbackground=GREEN,  
    font=("Consolas", 10),  
    relief="flat"  
)  
bob_private_entry.pack(pady=5)
```

```
bob_public_label = tk.Label(  
    bob_frame,  
    text="Public Key: ---",  
    font=("Consolas", 10),  
    bg="#101010",  
    fg=GREEN  
)  
bob_public_label.pack(pady=5)
```

```
bob_secret_label = tk.Label(  
    bob_frame,  
    text="Shared Secret: ---",  
    font=("Consolas", 10, "bold"),  
    bg="#101010",  
    fg=GREEN  
)  
bob_secret_label.pack(pady=5)
```

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```
button_frame = tk.Frame(root, bg=BLACK)
button_frame.pack(pady=5)
```

```
tk.Button(
    button_frame,
    text="[ EXCHANGE KEYS ]",
    command=calculate_keys,
    width=20,
    bg=DARK_GREEN,
    fg=GREEN,
    activebackground=GREEN,
    activeforeground=BLACK,
    font=("Consolas", 10, "bold"),
    relief="flat",
    cursor="hand2"
).grid(row=0, column=0, padx=8)
```

```
tk.Button(
    button_frame,
    text="[ CLEAR ]",
    command=clear_fields,
    width=12,
    bg="#151515",
    fg=WHITE,
    activebackground="#333333",
    font=("Consolas", 10, "bold"),
    relief="flat",
    cursor="hand2"
).grid(row=0, column=1, padx=8)
```

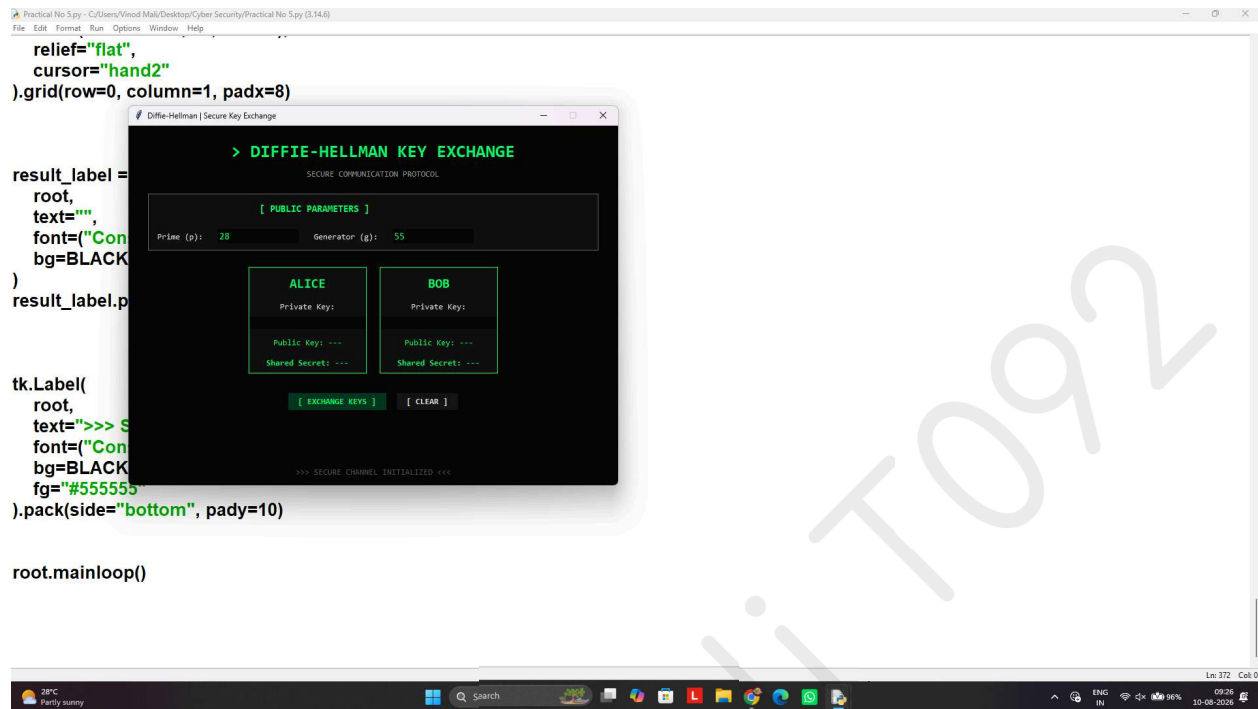
```
result_label = tk.Label(
    root,
    text="",
```

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```
font=("Consolas", 13, "bold"),  
bg=BLACK  
)  
result_label.pack(pady=15)  
  
tk.Label(  
    root,  
    text=">>> SECURE CHANNEL INITIALIZED <<<",  
    font=("Consolas", 9),  
    bg=BLACK,  
    fg="#555555"  
) .pack(side="bottom", pady=10)  
  
root.mainloop()
```

Output :-

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